CNN-Based Super-Resolution of Hyperspectral Images

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***Abstract*— Single-image super-resolution (SISR) techniques attempt to reconstruct the finer resolution version of a given image from its coarser version. In the SISR of hyperspectral data sets, the simultaneous consideration of spectral bands is crucial for ensuring the spectral fidelity. However, the high spectral resolution of these data sets affects the performance of conventional approaches. This research proposes the design of 3-D convolutional neural network (CNN)-based SISR architectures that can map the spatial–spectral characteristics of hypercubes to a finer spatial resolution. The proposed approaches facilitate the simultaneous optimization of sparse codes and dictionaries with regard to the super-resolution objective. Our main hypothesis is that the consideration of spectral aspects is essential for the spatial enhancement of hyperspectral images. Also, we propose that the regularized deconvolution of a coarser-scale hypercube, using learned 3-D filters, yields the required high-resolution version. Based on these hypotheses, a convolution–deconvolution framework is proposed to super-resolve the hypercubes in parallel with the reconstruction of a set of regularizing features. Novel sparse code optimization sub-networks proposed in this article give better performance than the existing strategies. The endmember similarities and hyperspectral image prior are considered while designing the proposed loss functions. In order to improve the generalizability, a collaborative spectral unmixing strategy is employed to refine the spectral base of the super-resolved result. The spatial–spectral accuracy of the super-resolved hypercubes, in terms of the validity of regularizing features and endmembers, is explored to devise an optimal ensemble strategy. The experiments, over different data sets, confirm better accuracy of the proposed frameworks compared to the prominent approaches.**

***Index Terms*— Alternating direction method of multiplier (ADMM), fast iterative shrinkage thresholding algorithm (FISTA), hyperspectral, super-resolution.**

I. Introduction

THE tradeoff between the spatial and spectral resolution of remote sensing satellite sensors results in mixed pixels in hyperspectral images. Single-image super-resolution (SISR) techniques, which attempt the reconstruction of finer resolution images from coarser ones, increase the probability of pure pixels. Unlike interpolation strategies, such as nearest neighbor, bicubic, and bilinear, super-resolution techniques can estimate the missing details to some extent. However, the prediction of subpixel values from coarse pixels and their neighborhood is a highly ill-posed problem. Prominent multi-source hyperspectral pan sharpening approaches [1]–[4] have also been widely explored to learn the spectral and spatial contexts from multi-source datasets. Sparse-coding-based SISR approaches, which learn low- and high-resolution dictionaries and sparse representations to transform coarser patches to finer resolution, have reported the state-of-the-art results. The basic hypothesis in these approaches is that an original sparse signal can be recovered from a set of measured values that are fewer than the propositions of the Shannon sampling theory. The iterative shrinkage thresholding algorithm (ISTA), which assumes the dictionaries to be learned at hand, iteratively approximates the sparse codes using a componentwise vector shrinkage function [5], [6]. A learned iterative version of ISTA, termed as LISTA, is proposed by Gregor and LeCun [7] in which a parameterized nonlinear encoder function is trained to predict the optimal sparse codes. In their seminal paper, Kamilov *et al.* [8] proposed a faster version of ISTA for total variation (TV)-based image denoising. Recently, deep convolutional architectures have been widely explored for learning the dictionaries and sparse codes [9]–[14]. Osendorfer *et al.* [15] showed that the upscaling in latent space yields better super-resolution reconstructions compared to the same performed in the image space. Dong *et al.* [9] illustrated that the joint enhancement of all image layers (Y, Cb, and Cr in YCbCr color space) yields better results than the conventional strategy of merely using the luminance (Y) component. Liu *et al.* [10] employed the domain expertise as a sparse prior to improve the deep-learning-based SISR approaches. Their paper discusses the implementation of a convolutional network whose layers correspond to different steps in the sparse-coding-based SISR pipeline. Wang *et al.* [11] proposed an ensemble of SISR models to avoid the possible convergence of stochastic gradient minimization toward local optima. However, the approach uses a backprojection method to compute the model weights, ignoring the spectral aspects of the data. Recently, Yang *et al.* [16] employed an alternating direction method of multiplier (ADMM)-based architecture for reconstructing finer scale magnetic resonance imaging (MRI) images from corresponding under-sampled versions.

Manuscript received September 28, 2018; revised June 5, 2019 and October 14, 2019; accepted February 7, 2020. Date of publication February 28, 2020; date of current version August 28, 2020. Arun P. V. and K. M. Buddhiraju acknowledge the funding support by the Department of Science and Technology, Government of India through the Network Project on Imaging Spectroscopy and Applications (NISA). This work was also supported in part by MIAI @ Grenoble Alpes, (ANR-19-P3IA-0003). *(Corresponding author: P. V. Arun.)*

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Digital Object Identifier 10.1109/TGRS.2020.2973370

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A majority of the existing SISR approaches give blurred and noisy results for hyperspectral images. Also, most of them do not simultaneously optimize the dictionaries and sparse codes with regard to the objective of super-resolution reconstruction. In addition, atmospheric effects and irregular topographies of aerial features make these approaches less effective. The model proposed by Osendorfer *et al.* [15], when implemented with more input bands, suffered a bottleneck because of ISTA. The prominent convolutional neural network (CNN)-based SISR models, such as those of Dong *et al.* [9], Liu *et al.* [10], Wang *et al.* [11], and Yang *et al.* [16], are found to be non-scalable with respect to an increase in the number of input spectral bands. A large spectral dimension increases the computational complexity and limits the learning/generalization capability of these models. The method of Liu *et al.* [10] suffers from uncertainty over the optimal network depth, and an increase in the number of network layers does not always guarantee better results. Shi *et al.* [17] illustrated the utility of deconvolution layers for SISR, but the approach is found to be unstable when tuned for hyperspectral data sets. The ADMM-based image reconstruction framework [16] results in overfitting for images with large spectral dimensions. Although the consideration of spectral aspects is critical for the proper super-resolution of hypercubes, most of the sparse-coding-based SISR methods ignore the same [18]. Recently, Yokoya *et al.* [19] and Lin *et al.* [20] adopted matrix factorization to extract a set of valid endmembers whose abundances served as the basis for enhancement. Similarly, Simões *et al.* [21] used endmember information to implement the enhancement in a low-dimensional space. However, these methods fuse the information from two or more images, and similar formulations experimented for sparse-coding-based hyperspectral SISR did not yield satisfactory results. In this research, a regularization mechanism, to bind all the spectral bands so as to project the spectral correlation to finer spatial scale, is investigated. Also, data-driven dimensionality reduction transformations are explored to address the issue of illposedness and to avoid the bottleneck of sparse code optimization strategies such as ISTA and ADMM.

Although not specifically for super-resolution, prominent CNN-based architectures available for similar image transformations are discussed in [22]–[27]. Recently, the concept of 3-D CNN has been proposed where the filters have a downward movement in addition to the movement in 2-D space [23], [28], [29]. Also, neural network inversion techniques [30]–[33], which attempt to reconstruct inputs from corresponding network outputs, are being widely employed for various image enhancement tasks [34]–[39]. In this article, different convolutional architectures as well as network inversion techniques are investigated to ensure spectral as well as spatial fidelity of the hyperspectral SISR. The proposed frameworks also facilitate the simultaneous optimization of sparse codes and dictionaries with regard to the super-resolution objective.

The CNN-based super-resolution techniques minimize the reconstruction loss, which is generally measured in terms of the mean squared error (MSE) between original and reconstructed images. However, these image space loss functions compute the mean of many possible solutions and cause artifacts, especially at high scale/zoom factors [40]. Recent studies have found that the loss computations in latent space improve the perceptual resemblance of the reconstructed images with respect to their natural versions [40], [41]. The generative adversarial network (GAN) [42], in which the generator network simultaneously learns to fool a discriminator, has been thoroughly explored for image enhancements [42], [43]. The use of conventional loss functions (perceptual as well as image space) for hyperspectral SISR causes artifacts and affects the spectral characteristics of the results. In this regard, this research investigates hypercube-specific loss regularizations for improving the spectral and spatial fidelity of the reconstructions.

Recently, neural network pruning has been widely employed to remove the least useful filters and channels from the network without affecting the accuracy. Generally, these approaches sparsify the filters and channels, and prune out the unimportant ones based on certain criteria. In this regard, low-rank matrix approximation [44]–[48], dynamic learning strategies [49]–[52], and structured sparse regularization [53]–[57] have been employed to sparsify a given network. He *et al.* [46] achieved the state-of-the-art results using a least absolute shrinkage and selection operator (LASSO)-regression-based channel and filter pruning. Kiaee *et al.* [58] proposed an ADMM-based approach to simultaneously induce sparsity and minimize the classification-based network loss. This research explores a few of these approaches, compatible with the 3-D CNNs, for pruning the proposed frameworks.

Based on the above discussions, the main contributions of this research can be summarized as follows.

1) 3-D CNN-based SISR architectures, which are capable of transforming the spatial–spectral characteristics of the hypercubes to a finer spatial scale.
2) Hypercube-specific loss functions to improve the learning capability of the network as well as the spectral fidelity of the super-resolved hypercubes.
3) Convolutional ISTA, fast iterative shrinkage thresholding algorithm (FISTA), and ADMM-based sparse code optimization strategies with learnable parameters.
4) Ensemble and network pruning strategies for hyperspectral SISR considering the spectral aspects of the data.
5) Collaborative unmixing-based spectral refinement to improve the generalization capability of the approach.

II. Proposed Approach

Consider a hyperspectral image/hypercube patch y^{lr} with spatial resolution s and number of spectral bands b. Let $y^{\mathrm{hr}\prime}$ be its high-resolution counterpart with a spatial resolution r. Here both r and s refer to the dimension on the ground of the fine and coarse resolution pixels, respectively, hence $r \ll s$. Reflectance of the mixed pixels in $\mathbf{y^{lr}}$ corresponds to the response from more than one class. Let N_{E} and $N_{\mathrm{E}'}$ be the number of endmembers in $\mathbf{y^{hr\prime}}$ and $\mathbf{y^{lr}}$, respectively. In this article, we attempt to recover a high-resolution patch $\mathbf{y^{hr}}$ (similar to the original ground truth $\mathbf{y^{hr\prime}}$) from its low-resolution version y^{lr}, ensuring the proper transformation of spectral properties. The following sections discuss the strategies adopted to formulate a CNN-based SISR framework to accomplish the same.

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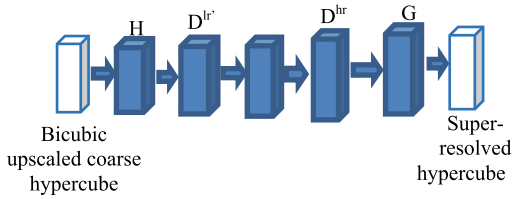


Fig. 1. Super-resolution architecture-1.

A. 3-D CNN-Based Architecture

In order to formulate an optimal network architecture for the spatial enhancement of hypercubes, a simple feedforward four-layer 3-D CNN (see Fig. 1) is first investigated. Unlike Liu *et al.* [10] and other related methods that employ 2-D filters and sparse codes, here 3-D filters are used. As illustrated in [18], [28], and [55], [59]–[61], the 3-D convolutions can better model the spectral aspects of the data when compared to their 2-D counterparts. In this article, the basic assumption is that the low-resolution and high-resolution image patches can be represented using overcomplete dictionaries ($\mathbf{D^{lr}}$ and $\mathbf{D^{hr}}$) along with some sparse coefficients (sparse codes, i.e., $\boldsymbol{\alpha}_{lr} = \boldsymbol{\alpha}_{hr} = \boldsymbol{\alpha}$). Thus, a high-resolution patch $\mathbf{y^{hr}}$ can be recovered from the corresponding low-resolution patch $\mathbf{y^{lr}}$ as

$$\begin{aligned} \mathbf{y}^{hr} &= \boldsymbol{\alpha} * \mathbf{D}^{hr} \\ \text{s.t. } \boldsymbol{\alpha} &= \arg\min_{Z} \left(\left\| \boldsymbol{y}^{lr} - \boldsymbol{Z} * \boldsymbol{D}^{lr} \right\|^2 + \lambda \|\boldsymbol{Z}\|_1 \right) \end{aligned} \tag{1}$$

where $*$ denotes the convolution operation. It may be noted that, in a backpropagation setting, (1) cannot be directly solved and hence is implemented indirectly. In this regard, $\mathbf{y^{lr}}$ is upscaled and projected onto the coarser dictionaries to yield the sparse codes ($\mathbf{Z}$), which are then subjected to nonlinear transformation using convolutional rectified linear units (ReLU) to enforce sparsity. The sparse codes are normalized so as to regulate their magnitude and are projected onto high-resolution dictionaries ($\mathbf{D^{hr}}$) to obtain the super-resolved output. The architecture in Fig. 1 (and other networks presented in this article) is tuned for training and testing hypercube patches having a spatial–spectral dimension of $64 \times 64 \times 172$. The blocks H and G denote patch generation and aggregation layers, while $\mathbf{D^{lr}}$ and $\mathbf{D^{hr}}$ denote the low- and high-resolution dictionaries, respectively. The layer H transforms the input image to overlapping patches, while layer G stitches the output patches to reconstruct the complete super-resolved image. Hence the proposed architecture can work with input images of any spatial dimension. The results of the experiments using standard hyperspectral data sets on the above network are found to be very poor. In most cases, when more than 20 spectral bands are used, acceptable results are not achieved even after fine-tuning the network parameters. The same experiments with fewer spectral bands yielded better results, suggesting that the learning capability is affected by an increase in the spectral dimension. In order to improve the results, the initial upscaling in architecture-1 (see Fig. 1) is replaced by a 3-D convolution–deconvolution network. The assumption is that the deconvolution of coarser scale images with learned 3-D filters can yield a spatially enhanced reconstruction of spectral bands. This approach removes the artifacts resulting from the use of static bi-cubic kernels

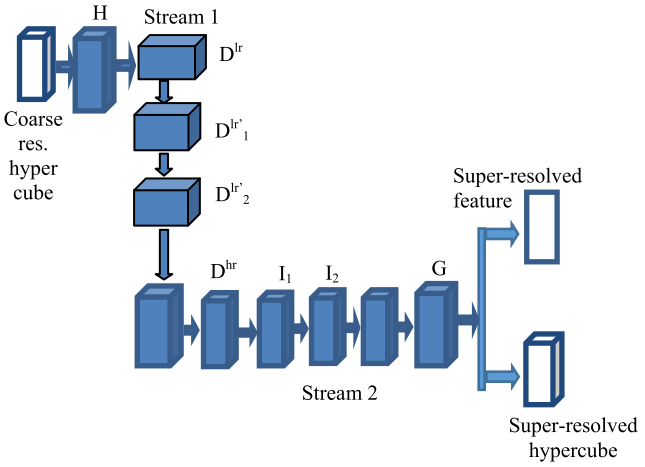


Fig. 2. Super-resolution architecture-2.

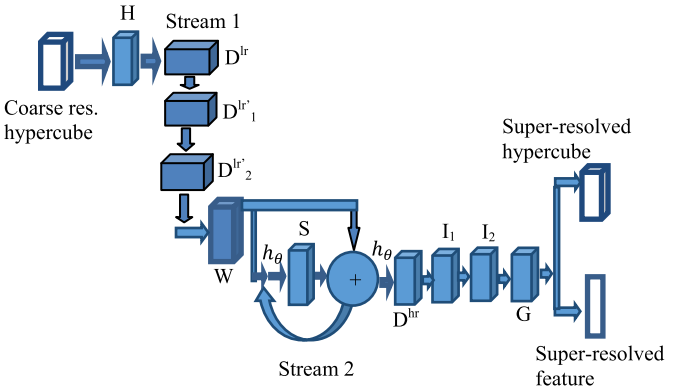


Fig. 3. Super-resolution architecture-3.

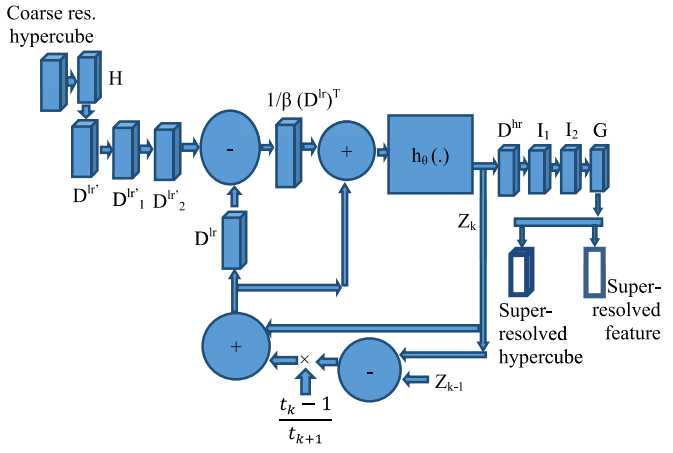


Fig. 4. Super-resolution architecture-4.

in the earlier approaches. Deconvolution is implemented as upscaling followed by convolution, where strides are decided based on the zoom factor, number of layers, and layerwise upscaling factors. For example, in the given architectures (see Figs. 2–6), the stride is set to one for the deconvolution layers as the layerwise upscaling = 2.

B. Feature-Guided Regularizations

Although dynamic upscaling is adopted, an increase in the spectral dimension of input impedes the simultaneous minimization of reconstruction losses of each band, thereby

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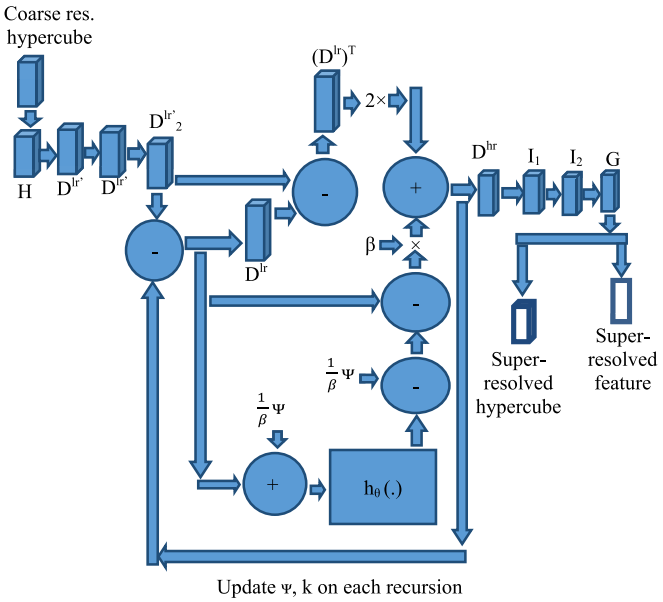


Fig. 5. ADMM-based super-resolution architecture-5.

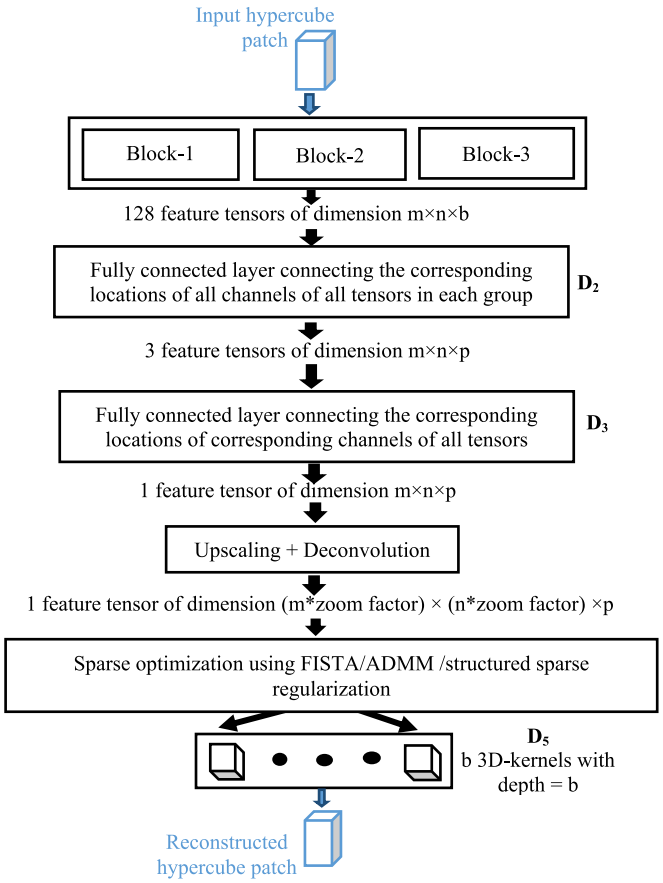


Fig. 6. Proposed structured encoding-based super-resolution architecture-6.

affecting the result. This issue is resolved by learning some features, derived from the entire hypercube, in parallel with the super-resolution reconstruction of the hypercube. The loss minimization of these features constrains the network to yield a hypercube from which such features can be generated. This approach ensures better spectral correlation and sharper reconstructions. In the current implementation, edges derived from the first principal component (PC) of the hypercube are used as the regularizing features. The refined architecture (architecture-2) is shown in Fig. 2. The stream-1 convolution–deconvolution network upscales the (coarse) input spectral bands to the desired spatial resolution. Stream-2 learns high-resolution dictionaries ($\mathbf{D^{hr}}$) and inverse mapping (I_1 and I_2) to generate the super-resolved hypercube along with regularizing features. The parametric ReLU block optimizes the upscaled sparse representation ($\mathbf{Z}$) generated at $\mathbf{D^{lr'}_2}$.

A reversible learning-based hyperparameter optimization, discussed in [62], is adopted to fine-tune the network parameters. Stream-1 has kernel sizes of 3×3 in the first, 3×3 in the second, and 7×7 in the third layer. The numbers of filters in the first, second, and third layers are set to be 516, 516, and 344, respectively. Similarly, the stream-2 convolution network uses kernel sizes of 1×1 in the first, 3×3 in the second, and 5×5 in the last three layers. The numbers of filters in the first three layers are set to 344, while the last two layers have 172 and 173 filters, respectively. The extra 3-D filter in the last layer is responsible for the generation of regularizing features, and its depth (third dimension of the kernel) is set to 172. The filter depth of all the other convolution and deconvolution units is set to 5.

Although architecture-2 has improved the results, the required optimal network depth is found to be sensitive to the data. Furthermore, the sparse codes are not necessarily sparse and mostly violate the basic assumptions [see (1)], resulting in corrupted dictionaries and reconstruction artifacts. These issues are due to the approximation of the sparse-code optimization process to a single-step convolution.

C. Sparse Code Optimization

In order to improve the accuracy and stabilize the results of architecture-2, the sparse representations ($\mathbf{Z}$) obtained at $\mathbf{D^{lr'}_2}$ need to be better optimized in a convolutional network setting as

$$\underset{\mathbf{Z}}{\operatorname{argmin}}\left(J(\mathbf{Z}) = \left(\left\| \boldsymbol{y}^{lr} - \mathbf{Z} * \boldsymbol{D}_2^{lr'} \right\|^2 + \lambda \|\mathbf{Z}\|_1\right)\right). \tag{2}$$

In this regard, Sections II-C1 and II-C2 explore the convolutional implementations of common proximal methods such as ISTA, FISTA, and ADMM for solving (2).

1) Convolutional Shrinkage Thresholding-Based Approach: This section discusses the optimization of $\mathbf{Z}$ in a convolutional network setting using the majorization–minimization algorithm and shrinkage thresholding operator. Employing the majorization–minimization algorithm, the next minimal point of $J(\mathbf{Z})$, $\mathbf{Z_{k+1}}$, is estimated by minimizing a function $G_k(\mathbf{Z})$ at $\mathbf{Z_k}$, such that $G(\mathbf{Z}) > J(\mathbf{Z})\ \forall\ \mathbf{Z} \neq \mathbf{Z_k}$ and $J(\mathbf{Z_k}) = G(\mathbf{Z_k})$. The function $G_k(\mathbf{Z})$, at each iteration k, is determined using the soft-thresholding operator [5], [6]. Hence, the sparse code optimization in (2) is iteratively formulated as

$$\mathbf{Z}_{k+1} = h_{\frac{\lambda}{2\beta}}(\mathbf{Z}_k * \boldsymbol{S} + \boldsymbol{y}^{lr'} * \boldsymbol{W}) \tag{3}$$

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where $\mathbf{S}$ and $\mathbf{W}$ are convolutional units, λ is the scaling constant, $\mathbf{y}^{\mathbf{lr}\prime}$ denotes the deconvolved features derived from $\mathbf{y}^{lr}$, $h_\theta(\cdot)$ is the shrinkage thresholding function with a threshold θ, and β is the step size. A proximal regularization strategy on the linearization of $J(\mathbf{Z})$ was also explored to reach an iterative formulation. However, (3) offers a comparatively simple architecture and, hence, is preferred. From (3), it is clear that the sparse code optimization can be accomplished using a recursive convolutional network. Assuming the threshold to be positive, the shrinkage thresholding function for the kth iteration $|h_\theta(a)|_k$ can be expressed as

$$|h_\theta(a)|_k = \text{sign}(a_k)\theta_k\left(\frac{|a_k|}{\theta_k} - 1\right) = \theta_k h_1\left(\frac{a_k}{\theta_k}\right). \quad (4)$$

According to (4), $h_\theta(a)$ can be implemented using two linear scaling layers (with weights θ and $1/\theta$, respectively) separated by a unit-thresholded ReLU layer. Based on (3) and (4), the sparse-code optimization is incorporated into the super-resolution pipeline as a recurrent sub-network, as shown in architecture-3 (see Fig. 3). It may be noted that, instead of using predefined thresholds (based on λ and β) like earlier approaches, here the network is made to learn the same. Architecture-3 has improved the learning capability and hence the reconstruction accuracy significantly. Furthermore, the results are found to be less sensitive to the network parameters.

Beck and Teboulle [63] proposed an improved version of ISTA called FISTA, where a very specific linear combination of the previous two points ($\mathbf{Z_k}$ and $\mathbf{Z_{k-1}}$) is considered for computing a given optimal point ($\mathbf{Z_{k+1}}$). The approach is observed to have a much faster convergence than ISTA. In a network setting, convolutional FISTA is iteratively formulated as

$$\mathbf{Z_{k+1}} = h_{\frac{\lambda}{\beta}}\left(\left(\mathbf{y}^{lr'} - \mathbf{U_k} * \mathbf{D}^{lr}\right) * \frac{1}{\beta}\mathbf{D}^{lr^{\mathrm{T}}} + \mathbf{U_k}\right) \quad (5)$$

$$\mathrm{t}_{k+1} = \frac{\left(1 + \sqrt{1 + 4t_k^2}\right)}{2} \quad (6)$$

$$\mathbf{U_{k+1}} = \mathbf{Z_k} + \frac{t_k - 1}{t_{k+1}}(\mathbf{Z_k} - \mathbf{Z_{k-1}}) \quad (7)$$

where $h_\theta(\cdot)$ is the soft-thresholding block and t_1 is set to 1. The proposed convolutional formulation of FISTA [see (5)–(7)], instead of ISTA, is embedded in architecture-3. The modified architecture (architecture-4) is presented in Fig. 4. The approach has significantly reduced the average number of recurrences required. Also, the network is much lighter when compared to the previous strategies. Moreover, the sensitivity of the approach toward the length of the recurrent network is reduced.

2) ADMM-Based Approach: In this section, instead of ISTA/FISTA, ADMM is explored in a CNN framework for optimizing the sparse codes. In this regard, (2) is reformulated as

$$\underset{Z}{\text{argmin}}(f(\mathbf{Z}) + \lambda R(\boldsymbol{\tau}))$$
$$\text{s.t. } \mathbf{Z} - \boldsymbol{\tau} = \mathbf{0}. \quad (8)$$

The formulation in (8) is clearly equivalent to (2) and can be decoupled into two parts that depend on two different variables. The augmented Lagrangian associated with the constrained problem in (8) is given as

$$C(\mathbf{Z}, \boldsymbol{\tau}, \Psi) = f(\mathbf{Z}) + \lambda R(\boldsymbol{\tau}) + \text{trace}\left(\Psi^{\mathrm{T}}(\mathbf{Z} - \boldsymbol{\tau})\right) + \frac{\beta}{2}\|\mathbf{Z} - \boldsymbol{\tau}\|_{\mathrm{F}}^2 \quad (9)$$

where Ψ is the dual variable matrix, β is a positive scalar, and $||\cdot||_{\mathrm{F}}$ is the Frobenius norm. A gradient descent-based approach is adopted to iteratively minimize (9) with respect to $\mathbf{Z}$. Hence for a given iteration k

$$\mathbf{Z_{k+1}} = \mathbf{Z_k} - \Big[2\left(\mathbf{Z_K} * D^{\mathrm{lr}} - \mathbf{y^{lr}}\right) * D^{\mathrm{lr}^{\mathrm{T}}} + \beta\left(\mathbf{Z_k} - \left(\boldsymbol{\tau_k} - \frac{1}{\beta}\Psi_\mathbf{k}\right)\right)\Big]. \quad (10)$$

Furthermore, $\mathbf{Z_{k+1}}$ and the soft-thresholding operator are employed to obtain $\boldsymbol{\tau_{k+1}}$, that is,

$$\boldsymbol{\tau_{k+1}} = \begin{cases} \left(1 - \frac{a}{\|\mathbf{V}\|_F}\right)\mathbf{V}, & \text{if } \|\mathbf{V}\|_F > a \\ \mathbf{0}, & \text{otherwise} \end{cases} \quad (11)$$

where $a = \lambda/\beta$, $\mathbf{V} = \mathbf{Z_{k+1}} + (1/\beta)\Psi_\mathbf{k}$. In the case of L_0 sparsity constraint, the update is formulated as

$$\boldsymbol{\tau_{k+1}} = \begin{cases} \mathbf{V}, & \text{if } \|\mathbf{V}\|_{\mathrm{F}} > b \\ \mathbf{0}, & \text{otherwise} \end{cases} \quad (12)$$

where $b = (2\lambda/\beta)^{1/2}$.

At each iteration, the dual variable Ψ is updated using a step-size equal to β as

$$\Psi_{\mathbf{k+1}} = \Psi_\mathbf{k} + \beta(\mathbf{Z_{k+1}} - \boldsymbol{\tau_{k+1}}). \quad (13)$$

From the experiments over standard data sets, the number of iterations required is found to be 20. The convolutional implementation of (10)–(13) for optimizing the sparse codes, in the hyperspectral SISR pipeline, is presented in Fig. 5. In this architecture, the dictionary block $\mathbf{D^{lr}}$ projects the upscaled latent representation of $\mathbf{y^{lr}}$ to a set of features that are iteratively optimized using the proposed ADMM sub-network. The block $h_\theta(\cdot)$ facilitates soft thresholding for a given threshold $\theta = \lambda/\beta$ so as to enforce the L_1 sparsity. As is evident, the approach simultaneously optimizes the sparse codes and dictionaries in an alternating manner. It may be noted that, unlike the ISTA-based strategies, the approach is more robust and better models the sparse coding problem preserving all the sparsity conditions. Also, it allows the flexibility of L_1 and L_0 sparsity constraints. However, the average number of recurrent cycles required for satisfactory results is much more compared to the FISTA.

D. Dimensionality Reduction

1) PC Transform-Based Approach: In order to avoid the bottleneck of FISTA/ISTA/ADMM due to the increased spectral dimension of the input hypercube, PCA-based dimensionality reduction is adopted. The input patches are transformed using PCA and the top p PC components are given as input to the framework instead of the entire b bands ($p \ll b$). The enhanced image in the PC domain is then inverted to

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obtain the final super-resolved result. It may be noted that the optimal number of PC components (value of p) for each patch is selected based on the number of endmembers in that patch. The vertex component analysis (VCA) [64] is used for estimating the number of endmembers. Although the approach resolves the bottleneck of the sparse-code optimization sub-networks, it results in reconstruction artifacts and affects the spectral fidelity of the results. This can be attributed to the sensitivity of the framework (particularly the optimal depth of the 3-D kernels) to the number of input bands.

2) Encoder-Based Approach: For resolving various issues related to the PCA-based strategy, an encoder-based SISR pipeline (architecture-6) is proposed (see Fig. 6). The approach implements data-driven transformation to a low-dimensional latent space before SISR. The layer-1 has three blocks among which block-1 consists of 16 groups of channelwise two-layer 2-D filters, while block-2 consists of 16 two-layer 3-D-filters with variable depths. Similarly, block-3 has 172 (since $b = 172$) 3-D filters (having depth $= b$) and an equal number of 2-D filters. It may be noted that, instead of pooling, here dilated convolutions [65] are adopted to facilitate proper modeling of the spatial contexts. The dilation rates are increased along each block. For instance, all the first 3-D filters in block-2 have a dilation rate of 1, while their corresponding second ones have a rate of 2. Layer-1 blocks the implement spectral–spatial convolutions resulting in 3-D tensor features having a dimension of $m \times n \times b$. These feature maps are grouped into three nonoverlapping groups based on the type of their respective blocks. A fully connected layer (level-2) is employed to merge and transform the tensors in each block to a reduced dimension of $m \times n \times p$ ($p \ll b$). The resulting three tensors are passed to the fully connected level-3 layer to obtain a single 3-D tensor feature, which is deconvolved to get the upscaled latent representation ($\mathbf{Z}'$). The $\mathbf{Z}'$, thus obtained, is subjected to FISTA/ADMM/ISTA-based optimization to obtain the sparse code ($\mathbf{Z}$), which is projected onto the level-5 high-resolution dictionary (consisting of b 3-D filters each having a depth p) to yield the super-resolved result. Instead of using the sparse code optimization sub-networks, a structured sparse regularization of filter weights and sparse codes was also experimented. Although the approach is computationally efficient, FISTA/ADMM-based strategy is found to be more effective and robust. In order to avoid overfitting, inspired from the idea of contractive autoencoders [66], Frobenius norm of the Jacobian matrix of the level-4 feature with respect to the input is minimized. The contractive regularization, thus formulated, is computed in a convolutional framework as

$$\boldsymbol{L_c} = \lambda_c \left\| \sum_{i=1}^{b} \left(\boldsymbol{Z} \odot \left(\left(\boldsymbol{J_{m,n,b}} - \boldsymbol{Z} \right) \odot \left(\boldsymbol{J_{m,n,b}} - \boldsymbol{Z} \right) \right) * \boldsymbol{D_i} \right) \right\|_{\mathrm{F}} \tag{14}$$

where λ_c is the scaling constant, $\mathbf{Z}$ is the output of the fourth layer, $\mathbf{D_i}$ denotes the ith filter at layer-5, $\mathbf{J_{m,n,p}}$ is the matrix of 1's having dimension $m \times n \times p$, $\odot$ denotes the Hadamard product, and $*$ denotes the convolution operation. It may be noted that L_c slows down the update of filter weights and thereby avoids overfitting. Compared to architectures 3–5, the recurrent length of the sparse code optimization sub-network of architecture-6 is considerably less.

E. Design of Loss Function

1) Proposed Loss Functions: The conventional loss functions such as MSE and inverse peak signal-to-noise ratio (PSNR) do not consider the spectral aspects of the data. Also, at higher scale factors, they result in blurred images with low spatial–spectral fidelity. In order to resolve these issues, the MSE loss function (L_{mse}) is supplemented with a hyperspectral loss (L_{hyper}), which is formulated as

$$L_{\mathrm{hyper}} = \lambda_{\mathrm{lc}} L_{\mathrm{content}} + \lambda_{\mathrm{ls}} L_{\mathrm{spectral}} + \lambda_{\mathrm{le}} L_{\mathrm{end}} \tag{15}$$

where λ_{mse}, λ_{lc}, λ_{ls}, and λ_{le} are normalizing constants. The content loss (L_{content}), between the reconstructed patch ($\mathbf{y^{hr}}$) and its corresponding ground truth ($\mathbf{y^{hr\prime}}$), is measured in a latent space as

$$L_{\mathrm{content}} = \frac{1}{M \cdot N \cdot P} \sum_{ijl} \left(\phi_k\left(\boldsymbol{y}^{\boldsymbol{hr}}_{ijl}\right) - \phi_k\left(\boldsymbol{y}^{\boldsymbol{hr}'}_{ijl}\right) \right)^2 \tag{16}$$

where $\phi_k(\cdot)$ denotes the kth convolution layer of the autoencoder, and M, N, P denote the dimensions of ϕ_k. Thus, L_{content} considers the structure, contrast, and gray level values while computing the similarity of image patches. In order to further ensure the desired spectral characteristics, the spectral-dissimilarity loss (L_{spectral}) is estimated as the inverse of the average spectral similarity between $\mathbf{y^{hr}}$ and $\mathbf{y^{hr\prime}}$, that is,

$$L_{\mathrm{specral}} = \frac{1}{M \cdot N} \left(\sum_{i=1}^{M} \sum_{j=1}^{N} \frac{\boldsymbol{y}^{\boldsymbol{hr}}{}_{ij} \cdot \boldsymbol{y}^{\boldsymbol{hr\prime}}{}_{ij}}{\left|\boldsymbol{y}^{\boldsymbol{hr}}{}_{ij}\right| \cdot \left|\boldsymbol{y}^{\boldsymbol{hr\prime}}_{ij}\right|} \right)^{-1} \tag{17}$$

where M, N are the dimensions of the reconstructed image patch and its corresponding ground truth. Although the use of L_{content} and L_{spectral} have given acceptable results, the network is further constrained to learn proper endmembers. The endmember loss (L_{end}) facilitates the network to favor solutions having valid endmembers and is defined as

$$L_{\mathrm{end}} = N_E \left(\sum_{j=1}^{N_{\mathrm{E}}} \frac{\boldsymbol{E}^{\boldsymbol{hr}}_{\boldsymbol{j}} \cdot \boldsymbol{E}^{\boldsymbol{hr\prime}}_{\boldsymbol{j}}}{\left|\boldsymbol{E}^{\boldsymbol{hr}}_{\boldsymbol{j}}\right| \cdot \left|\boldsymbol{E}^{\boldsymbol{hr\prime}}_{\boldsymbol{j}}\right|} \right)^{-1} \tag{18}$$

where $\boldsymbol{E}^{\boldsymbol{hr}}_{\boldsymbol{j}}$ and $\boldsymbol{E}^{\boldsymbol{hr\prime}}_{\boldsymbol{j}}$ are the jth endmembers derived from the super-resolved patch and its corresponding ground truth, respectively, and N_{E} is the number of endmembers derived using VCA. It may be noted that L_{spectral} and L_{end} improve the spectral as well as the endmember similarities between the original and reconstructed image patches.

2) Denoising Regularization: The hyperspectral images are prone to noise, and hence the proposed SISR frameworks adopt some regularizations for denoising the super-resolved result. In this regard, the TV regularizer [37], [67]–[70] is employed to ensure piecewise smoothness of the results. In other words, as the integral of the absolute gradient of a noisy image is high, reducing the TV along with the network loss facilitates denoising while preserving important

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details such as edges. As the spectral bands of a hypercube are supposed to be correlated, unlike the conventional approaches [67]–[69], the TV regularization adopted in this research minimizes the variations across both the spatial and spectral dimensions, that is,

$$R_{\varphi}(x) = \lambda_{\text{tv}} \sum_{i,j,k} \left(\left(\boldsymbol{y}_{i,j+1,k} - \boldsymbol{y}_{ijk} \right)^2 + \left(\boldsymbol{y}_{i+1,j,k} - \boldsymbol{y}_{ijk} \right)^2 + \left(\boldsymbol{y}_{i,j,k+1} - \boldsymbol{y}_{ijk} \right)^2 \right)^{\frac{2}{\varphi}} \quad (19)$$

where R_{φ} is the TV regularizer, λ_{TV} denotes the normalizing constant, and φ is a scalar whose value determines the tradeoff between denoising and sharpness preservation. In order to reduce the computational complexity, instead of using the entire hypercube for TV computation, the top three PCs are used. Although TV loss is differentiable and can be minimized along with the network loss using gradient descent, a GAN-based implementation is experimented where the TV values are used as the discriminator criteria. An advantage of the GAN-based approach is that an additional spectral–spatial regularizer such as endmember thresholding can be used along with the TV loss without affecting the solvability of the optimization problem.

F. Design of an Ensemble Strategy

The convergence of stochastic gradient-based loss minimization (adopted in the proposed models) depends on the network parameter settings as well as initializations. Furthermore, high correlation among the spectral bands along with the interband noise affects the robustness of the proposed models. For resolving these issues, an ensemble of the proposed framework is implemented by employing the models with different initializations/parameter settings; the final super-resolved image is derived as a (normalized) weighted combination of the outputs of all the constituent models. Endmember similarities as well as the resemblance between the regularizing features derived from the input and the downscaled version of the output are considered for computing the model weights, that is,

$$\text{W}_i = \frac{1}{\left\| \boldsymbol{F}_{\boldsymbol{input}} - \boldsymbol{F}'_{\boldsymbol{oi}} \right\|_2} + \frac{1}{N_{\text{E}}} \sum_{j=1}^{N_{\text{E}}} \frac{\boldsymbol{E}_{\boldsymbol{j}}^{\boldsymbol{lr}} \cdot \boldsymbol{E}_{\boldsymbol{ij}}^{\boldsymbol{hr}'}}{\left| \boldsymbol{E}_{\boldsymbol{j}}^{\boldsymbol{lr}} \right| \cdot \left| \boldsymbol{E}_{\boldsymbol{ij}}^{\boldsymbol{hr}'} \right|} \quad (20)$$

where W_i is the weight of the ith model, F_{input} and F'_{oi} are the features derived from the input and downscaled output of the ith model, respectively, $\mathbf{E}_j^{lr}$ and $\mathbf{E}_{ij}^{hr\prime}$ are the jth endmembers derived from the input and the downscaled output of the ith model, respectively, and N_{E} is the number of endmembers. The use of feature as well as endmember similarity for weight computation reduces the computational complexity and results in outputs having better spectral correlation.

G. Design of Network Pruning Strategy

This section discusses a few of the approaches adopted for pruning the proposed networks. The multidimensional nature of 3-D convolutional blocks as well as the tradeoff between the computation time and generalization capability of the network are considered while formulating the pruning strategy. Initially, a subset of the training data is employed to train the network so as to sparsify a set of filters and channels that are redundant or not useful with regard to the given data. In this regard, structured sparse regularizations, such as the ones employed for channel and filter sparsifications in architecture-6, are adopted. Further to that, criteria such as L_0 norm of the filters and channels, L_0 norm and standard deviation of activation maps, and mutual information between the activation maps and network outputs are employed for ascertaining the importance of filters. The unimportant filters, based on a threshold, are removed. Finally, the pruned network is retrained to fine-tune the filter weights. The three stages are repeated for a predecided number of iterations. A normalizing constant (λ_p) is employed for deciding the tradeoff between pruning regularizations and reconstruction loss functions.

H. Coarser Scale Spectral Prior-Based Refinement to Improve the Generalization Capability

In order to fine-tune the super-resolved result ($\mathbf{y}^{\mathbf{hr}}$) with respect to the coarser input image ($\mathbf{y}^{\mathbf{lr}}$), a collaborative spectral unmixing [19], [71]-based optimization strategy is proposed. The following multiplicative update rules are employed for implementing the joint unmixing:

$$\boldsymbol{E} = \boldsymbol{E} \cdot * \left(\boldsymbol{y}^{\boldsymbol{lr}} \boldsymbol{A}^{\boldsymbol{T}} \right) ./ \left(\boldsymbol{E} \boldsymbol{A} \boldsymbol{A}^{\boldsymbol{T}} \right) \quad (21)$$

$$\boldsymbol{A} = \boldsymbol{A} \cdot * \left(\boldsymbol{E}^{\boldsymbol{T}} \boldsymbol{y}^{\boldsymbol{lr}} \right) ./ \left(\boldsymbol{E}^{\boldsymbol{T}} \boldsymbol{E} \boldsymbol{A} \right) \quad (22)$$

$$\boldsymbol{E}' = \boldsymbol{E}' \cdot * \left(\boldsymbol{y}^{\boldsymbol{hr}} \boldsymbol{A}'^{\boldsymbol{T}} \right) \cdot / \left(\boldsymbol{E} \boldsymbol{A}' \boldsymbol{A}'^{\boldsymbol{T}} \right) \quad (23)$$

$$\boldsymbol{A}' = \boldsymbol{A}' \cdot * \left(\boldsymbol{E}'^{\boldsymbol{T}} \boldsymbol{y}^{\boldsymbol{hr}} \right) \cdot / \left(\boldsymbol{E}'^{\boldsymbol{T}} \boldsymbol{E}' \boldsymbol{A}' \right) \quad (24)$$

where $\mathbf{E}$ and $\mathbf{A}$ are the end member and abundance matrices respectively of $\mathbf{y}^{\mathbf{lr}}$, and $\mathbf{E}'$ and $\mathbf{A}'$ are the endmember and abundance matrices, respectively of $\mathbf{y}^{\mathbf{hr}}$. The additive (sum to one) constraint is enforced using the approach discussed in Heinz and Chang [72]. For unmixing $\mathbf{y}^{\mathbf{lr}}$, the endmember matrix $\mathbf{E}$ is initialized using the endmember information derived from $\mathbf{y}^{\mathbf{lr}}$. The abundance matrix A is set to a constant value of $(1/N_{\text{E}})$ and is updated using (21) until convergence, keeping the value of E fixed. The $\mathbf{A}$ and $\mathbf{E}$ are then optimized by alternately applying (21) and (22). For unmixing $\mathbf{y}^{\mathbf{hr}}$, $\mathbf{E}'$ is initialized using $\mathbf{E}$ and $\mathbf{A}'$ is updated using (24). The update rules (23) and (24) are employed alternately for optimizing $\mathbf{A}'$ and $\mathbf{E}'$. The two unmixing steps are then repeated alternately with each time $\mathbf{E}$ initialized using $\mathbf{E}'$ for the unmixing of $\mathbf{y}^{\mathbf{lr}}$. For each iteration, $\mathbf{E}$, $\mathbf{E}'$, $\mathbf{A}$ and $\mathbf{A}'$ are updated using (21)–(24). The change in value of the cost function between two successive iterations is checked to be below a small threshold (ε) as the condition for convergence. When the number of iterations exceeds a predefined maximum or when the convergence is achieved, the final super-resolved image is computed as

$$\boldsymbol{y}_{\boldsymbol{f}}^{\boldsymbol{hr}} = \boldsymbol{E}'' \boldsymbol{A}'' \quad (25)$$

where $\mathbf{E}''$ and $\mathbf{A}''$ are, respectively, the updated version of $\mathbf{E}'$ and $\mathbf{A}'$. It may be noted that the spectral prior-based transformation fine-tunes the super-resolved result by aligning the spectral base toward the coarse resolution input. Although a sparse prior-based transformation was also experimented, this approach is found to be better.

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III. Experiments

Different super-resolution architectures proposed in this article are analyzed over various scale/zoom factors (2, 3, and 4). Due to the absence of benchmark hyperspectral data for evaluating the proposed frameworks, different standard AVIRIS hyperspectral images along with those collected from known locations are used for training and testing the networks. During training, the mean per channel calculated over all the images is subtracted from the images. The training set employed in this article consists of 5000 patches, and the models are tested on a set of ten images (1000 patches). The digital numbers of all the images are mapped to the range of [0−1]. The high-resolution image patches are down-scaled using different interpolation strategies such as bilinear, bi-cubic, and nearest neighbor interpolation to generate high- and low-resolution training/testing pairs. The training data are augmented with translation, rotation, scaling, Gaussian noise, blurring (Gaussian blur and smoothing), and advanced GAN-based augmentations. For the experiments to be reproducible and comparable, the results presented in this article are for the standard hyperspectral data sets. Throughout the experiments, the mini-batch size for training is set to 200; momentum and weight decay for the backpropagations are set to 0.8 and 10^{-3} respectively (obtained through cross-validation); learning rate is initially set to 0.6 and is depreciated by a factor of 10 after every 100 epochs. All the models are analyzed for 200 epochs. The PSNR, spectral similarity measure (SSM), and structural similarity index measure (SSIM), computed between the reconstructed images and their corresponding high-resolution counterparts, are used as quantitative measures for estimating the accuracy. PSNR does not accurately represent the human perception; hence SSIM as well as classification accuracies (kappa statistics) are also reported for better comparison. The kappa statistics is computed over the classification results generated from the reconstructed images.

TABLE I
Comparative Analysis of the Proposed Architectures

Architecture	Zoom factor	Avg. PSNR	Avg. SSM	Avg. Kappa	Avg. Comp. time (s)
Architecture-1	3	26.42	0.9778	0.79	279
	4	23.09	0.9725	0.71	354
Architecture-2	3	28.62	0.9938	0.83	328
	4	26.84	0.9912	0.80	393
Architecture-3	3	33.62	0.9965	0.94	489
	4	30.71	0.9951	0.90	568
Architecture-4	3	34.19	0.9984	0.94	545
	4	33.02	0.9978	0.90	639
Architecture-5	3	36.72	0.9984	0.94	592
	4	34.98	0.9970	0.91	706
PCA based dimensionality reduction	3	32.71	0.9893	0.92	182
	4	30.59	0.9825	0.89	259
Architecture-6	**3**	**38.42**	**0.9991**	**0.96**	**283**
	4	**36.09**	**0.9986**	**0.92**	**368**
Proposed network pruning	3	35.72	0.9918	0.92	135
	4	34.18	0.9890	0.89	218

A. Study of Architectures

A comparative analysis of the proposed architectures over Indian pines, Pavia, and KSC data sets (for zoom factors 3 and 4) is presented in Table I. As is evident from Tables I, IV, and V, the architecture by Liu *et al.* [10] did not yield satisfactory results for experiments with 172 input bands (joint band enhancement). A similar model with 3-D CNN and inverse feature mapping (i.e., architecture-1) has given better reconstructions. The diversity and stability of the learned filters have been significantly improved for architecture-2. However, the feature maps in stream-2 did not seem to comply with the basic sparse code assumptions, indicating the inability in learning the sparse codes. Embedding of the sparse code optimization sub-network for resolving this issue, as explained in architectures 3–6, has resulted in better and stable reconstructions at fewer backpropagations, even for higher zoom factors. The experiments indicate that the proposed FISTA and ADMM-based optimization sub-networks have yielded significant improvements when compared to the ISTA-based approach. Among the two, the ADMM-based one has yielded slightly better results, but is computationally more complex. Also, it has illustrated better generalization capability and should be preferred when the training samples are less. Compared to ADMM, ISTA requires much smaller numbers of recurrent cycles for reasonable results. Although the reduction in dimensionality using PCA improves the computational efficiency, the approach results in reconstruction artifacts. These issues are resolved by architecture-6, which significantly improves the results and avoids optimization bottlenecks. The sparse-coding-based network pruning has reduced the computational complexity many folds, however, the same has slightly affected the reconstruction accuracies.

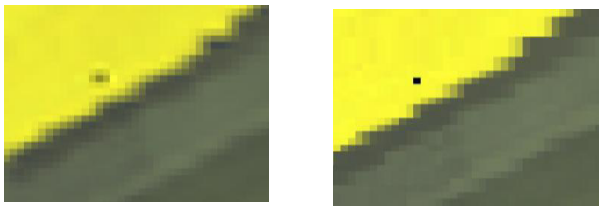
B. Analysis of Loss Functions

A summary of the ablation analysis of loss functions is presented in Table II. As is evident, the proposed loss functions yield better results compared to their conventional counterparts. The content loss (L_{content}), computed in the perceptual domain, gives better fidelity (natural resemblance) when compared to the image space-based loss functions. However, for improved perceptual resemblance and higher PSNR, both the losses are used together. The effects of the proposed losses are better captured using SSIM and SSM rather than with MSE/PSNR. It may be noted that, unlike in earlier approaches, L_{content} is computed using the features derived from the hyperspectral reconstruction network and not from the Visual Geometry Group (VGG) network [73]. Unlike the MSE-based dissimilarity loss, which computes the MSE between corresponding bands and averages them, L_{content} estimates the error between features, each of which is derived from the entire hypercube. Hence, L_{content} significantly improves the reconstruction, especially at higher scale factors. The improvement is more significant as the spectral dimension

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TABLE II
COMPARATIVE ANALYSIS OF LOSS FUNCTIONS

Loss	Zoom factor	Avg. PSNR	Avg. SSM
L_{MSE}	3	29.93	0.9844
	4	27.81	0.9716
$L_{content}$	3	28.34	0.9899
	4	25.96	0.9790
$\lambda_M L_{mse} + \lambda_{lc} L_{content}$	3	32.77	0.9911
	4	28.38	0.9838
$\lambda_{mse} L_{mse} + \lambda_{ls} L_{spectral}$	3	34.69	0.9925
	4	31.05	0.9860
$\lambda_{mse} L_{mse} + \lambda_{le} L_{end}$	3	33.54	0.9916
	4	29.85	0.9890
$\lambda_{mse} L_{mse} + \lambda_{lc} L_{content} + \lambda_{ls} L_{spectral}$	3	36.23	0.9925
	4	33.67	0.9962
$\lambda_{mse} L_{mse} + \lambda_{lc} L_{content} + \lambda_{le} L_{end}$	3	34.91	0.9937
	4	32.78	0.9959
$\lambda_{mse} L_{mse} + \lambda_{lc} L_{content} + \lambda_{le} L_{end} + \lambda_{ls} L_{spectral} + \lambda_{tv} L_{tv}$	**3**	**38.42**	**0.9991**
	4	**36.09**	**0.9986**



(a) (b)

Fig. 7. Illustration of the effect of the proposed approach in artifacts removal at zoom factor = 4. (a) Artifacts due to MSE losses. (b) Result of proposed losses

of the input increases. Also, L_{content}, computed based on the spectral–spatial features of the entire hypercube, does not cause much distortion effect, which is seen in other conventional networks.

As is evident from Fig. 7, the spectral-dissimilarity loss (L_{spectral}) and L_{content}, presented in this article, improve the reconstructions when compared to the conventional approaches (GAN and VGG). The proposed spectral losses (L_{spectral} and L_{end}) are easily computable and they ensure proper transformation of the spectral features to a finer spatial scale. Also, the use of TV regularizer has significantly improved the results, particularly for noisy data. It may be noted that a Gaussian noise-based data augmentation is employed to train and test the denoising regularization. Based on a grid search-based optimization and cross-validation over various data sets, the optimal values of normalizing constants λ_{mse}, λ_{lc}, λ_{ls}, and λ_{le} are found to be 0.450, 0.270, 0.060, and 0.040, respectively. For TV regularization, optimal values for φ and normalizing constant (λ_{tv}) are observed to be 2 and 0.18, respectively.

The considerable impact of the proposed loss functions on the performance of the approach can mainly be attributed to the consideration of the entire hypercube as an entity for the computation of these losses. It may be noted that L_{content}, L_{spectral}, and L_{end} consider the hypercube as a whole rather than considering it as a stack of independent spectral bands. At lower spectral dimension, the difference between the proposed loss functions and the MSE-based ones is not significant.

C. Analysis of Network Parameter Settings

Among the proposed CNN-based SISR frameworks, the performance as well as the accuracy of architecture-1 is highly sensitive to the network depth. Although sensitivity to the network depth has been considerably reduced for architecture-2, the selection of an optimal depth for stream-2 is still ambiguous. The architectures 3–6 provide stable results and are comparatively less sensitive to the number of layers. This can be attributed to the stability achieved through sparse code optimization. The number of recurrence cycles of the optimization sub-network is empirically set to ten for architectures 3 and 4, and minor variations in the same do not affect the reconstruction. The sensitivity toward the length of the recurrence cycle is the least for architectures 4–6. The variation in reconstruction PSNR with respect to the depth of different network streams for architecture-5 is illustrated in Fig. 8. It can be observed that there is only a slight rise in PSNR even for a significant increase in the network depth. Also, the trend is found to slowly saturate. Moreover, an increase in the number of layers (convolution units) increases the computational cost of training. Hence a better tradeoff is needed.

The increase in the number of filters in each layer generally improves the PSNR but increases the running time. Hence, an optimal choice depends on the tradeoff between both. For architectures 1–5, the increase in size of the convolution filters improves the PSNR only to a certain limit. An analysis of the trend reveals that the blind increase may affect not only the computational performance but also the PSNR, owing to the inability of the network to learn smaller features. However, increase in the size of deconvolution filters generally improves the results, but at the cost of running time. Hence, the optimal filter sizes should be determined with reference to the scene homogeneity/complexity. Empirically, kernel sizes of 3–7 are found to be optimal for most of the data sets. The optimal depth of the filters depends on the number of spectral bands as well as the spectral correlation. Hence the kernel depths need to be tuned with respect to the training data. It is further observed that increase in the diversity of kernel sizes improves the accuracy and yields better reconstructions even at shorter depths.

D. Effect of Regularizing Features

An analysis of the effectiveness of the proposed feature reconstruction regularization is summarized in Table III. As is evident, the use of feature reconstruction loss along with the regular hypercube reconstruction loss yields better results. This can be attributed to the ability of the former in binding the reconstruction of individual bands. Empirically, the values of

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TABLE III
COMPARATIVE ANALYSIS OF THE PROPOSED ARCHITECTURES

Regularizing features	Zoom factor	Avg. PSNR	Avg. SSM	Avg. Kappa
Reconstruction loss	3	35.21	0.9982	0.94
	4	33.87	0.9935	0.89
λ_{recon} Reconstruction loss+ $\lambda_{feature_recon}$ Feature reconstruction loss	**3**	**38.42**	**0.9991**	**0.96**
	4	**36.09**	**0.9986**	**0.92**

λ_{recon} and $\lambda_{feature_recon}$ are found to be 0.82 and 0.18, respectively. The use of edges as regularizing features facilitates sharper reconstructions. In order to analyze the effect of the regularizing features on the proposed model's accuracy, edges at different orientations (derived from the PCs) are employed. For all the architectures, increase in the number of regularizing features improves the accuracy, but at the cost of increased running time. This can be attributed to the enhanced cross-spectral learning. The edges derived from the first three PCs (of the hypercube) have yielded better results when compared to those derived from the first PC. The number of filters, as well as the kernel diversity, may be increased to maintain the pace of these improvements.

E. Evaluation of the Proposed Method

The PSNR, SSM, kappa, and SSIM values estimated for different approaches, on standard data sets, are presented in Table IV. Many of the state-of-the-art super-resolution techniques are found to be not suitable for hyperspectral images. In this regard, individual band enhancements (Bicubic, [9]–[11]) along with some recent hyperspectral SISR approaches [18], [74]–[77] are used as benchmarks for evaluating the proposed frameworks. For the SISR method of Wang *et al.* [11], the number of models in the ensemble is set to four. From the results (see Tables II and III), it can be observed that the proposed architectures (especially architectures 4–6) outperform the other methods. The individual band enhancements have produced poor results (particularly, in terms of SSM) when compared to the approaches that consider simultaneous/joint enhancement of all bands. For prominent SISR approaches such as those of Dong *et al.* [9], Liu *et al.* [10], and Wang *et al.* [11], an increase in the spectral dimension of the input affects the spectral as well as spatial fidelity of the results. The regularization proposed in our approach successfully resolves this issue, making the network scalable with respect to the spectral dimension of input. The proposed loss functions improve the perceptual and spectral accuracies, which is evident from the improved kappa statistics, SSIM, and SSM values. As illustrated in Tables II and III and Figs. 9 and 10, the use of deconvolution units along with recurrent sub-networks removes artifacts and considerably reduces the reconstruction error. Analysis of the variation in the accuracy and performance with respect to the ensemble size (number of models) indicates that increase in the number of ensemble models, although it increases the running time, improves accuracy. Recent

TABLE IV
COMPARATIVE ANALYSIS OF THE PROPOSED METHOD WITH PROMINENT SISR APPROACHES FOR ZOOM FACTOR = 3

Dataset	Approach	PSNR	SSIM	SSM	Kappa
Pavia	Feng et al. [75]	28.14	0.664	0.9264	0.81
	Liu et al. [10]	26.32	0.638	0.8618	0.76
	Wang et al. [11]	27.09	0.652	0.8691	0.78
	Dong et al. [74]	28.98	0.706	0.9295	0.83
	Li et al. [76]	29.66	0.740	0.9528	0.85
	Fu et al. [77]	30.84	0.763	0.9654	0.88
	Mei et al. [18]	31.49	0.779	0.9802	0.90
	Proposed architecture-5	34.07	0.835	0.9871	0.93
	Proposed architecture-6	**35.23**	**0.881**	**0.9916**	**0.94**
Indian Pines	Feng et al. [75]	30.09	0.682	0.9291	0.83
	Liu et al. [10]	28.14	0.651	0.8639	0.76
	Wang et al. [11]	29.02	0.679	0.8876	0.79
	Dong et al. [74]	30.87	0.721	0.9365	0.84
	Li et al. [76]	31.65	0.765	0.9591	0.87
	Fu et al. [77]	32.12	0.781	0.9704	0.89
	Mei et al. [18]	32.98	0.807	0.9826	0.91
	Proposed architecture-5	35.74	0.842	0.9873	0.96
	Proposed architecture-6	**36.92**	**0.890**	**0.9968**	**0.96**
KSC	Feng et al. [75]	30.82	0.749	0.9339	0.84
	Liu et al. [10]	29.87	0.698	0.8743	0.77
	Wang et al. [11]	29.53	0.704	0.8978	0.80
	Dong et al. [74]	31.21	0.736	0.9415	0.85
	Li et al. [76]	31.83	0.752	0.9665	0.89
	Fu et al. [77]	32.46	0.793	0.9731	0.92
	Mei et al. [18]	34.90	0.822	0.9807	0.94
	Proposed architecture-5	36.87	0.869	0.9894	0.97
	Proposed architecture-6	**37.59**	**0.894**	**0.9940**	**0.97**
Salinas	Feng et al. [75]	32.90	0.786	0.9408	0.86
	Liu et al. [10]	30.46	0.712	0.8954	0.80
	Wang et al. [11]	31.79	0.736	0.9130	0.82
	Dong et al. [74]	32.85	0.754	0.9597	0.86
	Li et al. [76]	33.58	0.790	0.9689	0.90
	Fu et al. [77]	35.02	0.819	0.9782	0.92
	Mei et al. [18]	36.57	0.843	0.9896	0.94
	Proposed architecture-5	38.32	0.881	0.9903	0.97
	Proposed architecture-6	**39.94**	**0.913**	**0.9961**	**0.98**

SISR approaches [9]–[11], [15] adopt cascading of networks for higher zoom factors. However, the dynamic upscaling, implemented in stream-1 layers, gives better results. Furthermore, for hyperspectral data, the cascade approach is computationally too complex to be practical.

Although the comparative analyses discussed above illustrate better accuracy and performance of the proposed approaches, the use of a limited number of standard data sets affects the effectiveness and reliability of such comparisons. For a fair evaluation, the networks have been trained and tested with mutually exclusive UAV and AVIRIS-NG image patches. The results obtained for the latter are presented in Fig. 11. Both these experiments over 18 030 training patches and 80 test images indicate that the proposed approaches

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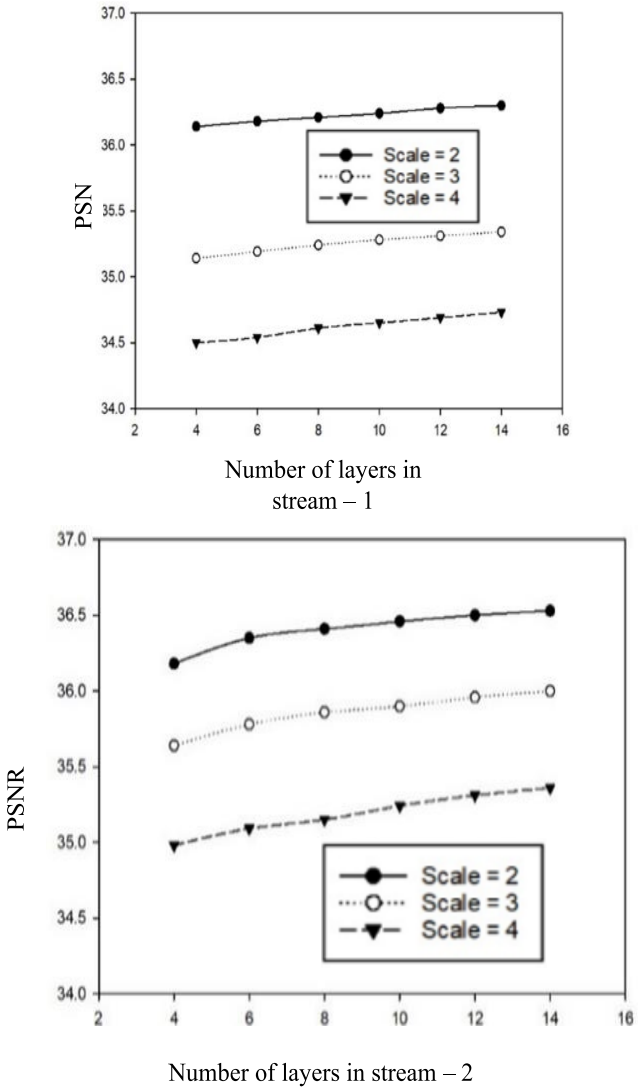


Fig. 8. Variation in reconstruction PSNR with stream-1 and 2 network-depths for architecture-5.

perform better than the prominent hyperspectral SISR techniques. As illustrated in Fig. 12, the proposed approach gives results comparable to the state-of-the-art even when tuned for RGB images. In order to further assess the robustness of the approach, one among the standard data sets was excluded from the training set and was used for testing. For instance, the *Indian Pines* image is tested on a network trained with the patches of Salinas, Pavia, and KSC. The results are summarized in Table V. As is evident, the proposed approaches have given significant improvement when compared to the other approaches. This can be attributed to the proposed sparse prior mapping as well as the spectral unmixing-based refinement. For analyzing the noise resilience of the approach, additive Gaussian noise corrupted input coarse images have been employed. The proposed approach is found to be noise resilient to some extent, mainly due to the use of TV regularizer as well as the regularizing features. The experiments indicate that Gaussian noise-based augmentation of the training data sets further improves the results.

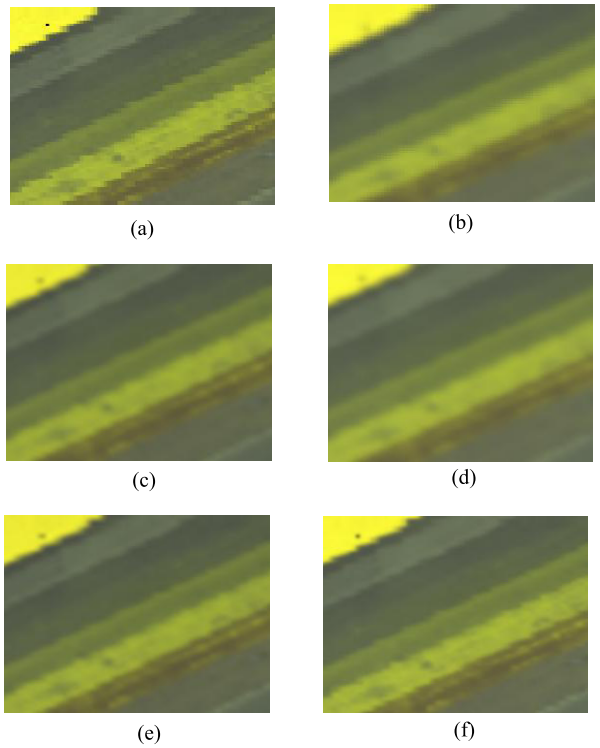


Fig. 9. Results of the proposed approach on Salinas data set for zoom factor = 4. (a) High-resolution image. (b) Simulated coarser version. (c) Dong *et al.* [74]. (d) Fu *et al.* [77]. (e) Mei *et al.* [18]. (f) Proposed approach.

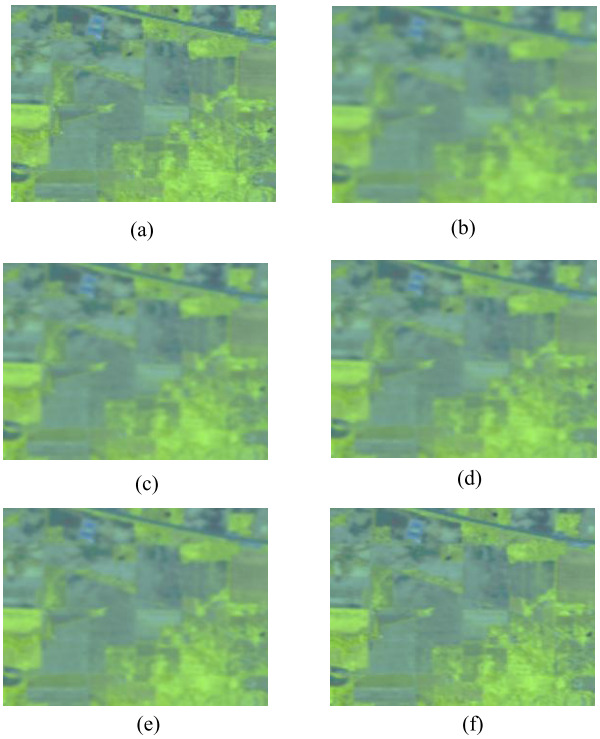


Fig. 10. Results of the proposed approach on Indian Pines data set for zoom factor = 4. (a) High-resolution image. (b) Simulated coarser version. (c) Dong *et al.* [74]. (d) Fu *et al.* [77]. (e) Mei *et al.* [18]. (f) Proposed approach.

In the above discussed experiments, a simulated coarse image downscaled from the original image is used as the input. The reconstructed image is then compared with the

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TABLE V
COMPARATIVE ANALYSIS OF THE PROPOSED METHOD WITH PROMINENT SISR APPROACHES FOR ZOOM FACTOR = 3

Dataset	Approach	PSNR	Kappa
KSC	Feng et al. [75]	20.32	0.74
	Dong et al. [74]	22.60	0.76
	Li et al. [76]	23.95	0.76
	Fu et al. [77]	26.41	0.79
	Mei et al. [18]	27.09	0.81
	Proposed architecture-5	28.73	0.83
	Proposed architecture-6	**29.92**	**0.86**
Salinas	Feng et al. [75]	23.12	0.75
	Dong et al. [74]	24.73	0.77
	Li et al. [76]	27.38	0.81
	Fu et al. [77]	28.05	0.82
	Mei et al. [18]	28.92	0.84
	Proposed architecture-5	30.86	0.87
	Proposed architecture-6	**31.14**	**0.90**
Pavia	Feng et al. [75]	19.89	0.72
	Dong et al. [74]	21.16	0.73
	Li et al. [76]	21.98	0.75
	Fu et al. [77]	22.81	0.78
	Mei et al. [18]	24.13	0.80
	Proposed architecture-5	27.94	0.83
	Proposed architecture-6	**28.37**	**0.84**
Indian Pines	Feng et al. [75]	21.68	0.74
	Dong et al. [74]	23.40	0.76
	Li et al. [76]	24.76	0.78
	Fu et al. [77]	26.95	0.79
	Mei et al. [18]	27.88	0.81
	Proposed architecture-5	29.32	0.84
	Proposed architecture-6	**30.93**	**0.87**

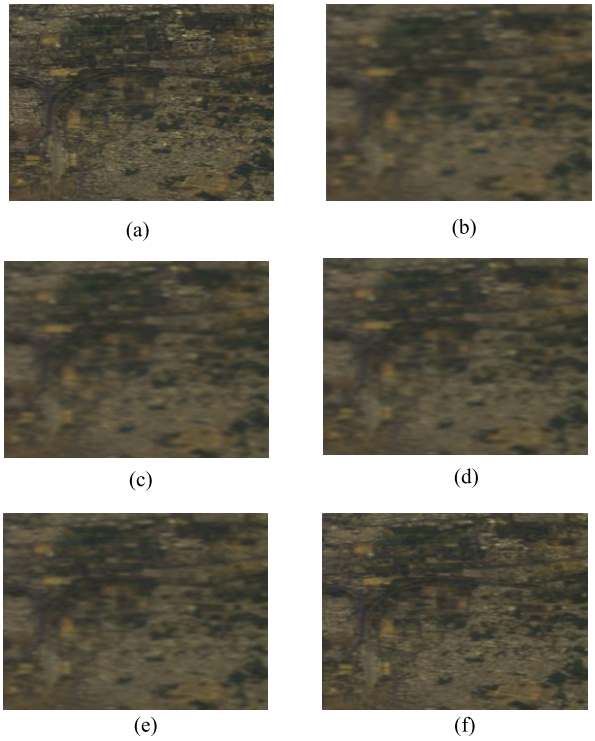


Fig. 11. Results of the proposed approach on AVIRIS-NG data set for zoom factor = 4. (a) High-resolution image. (b) Simulated coarser version. (c) Dong *et al.* [74]. (d) Fu *et al.* [77]. (e) Mei *et al.* [18]. (f) Proposed approach.

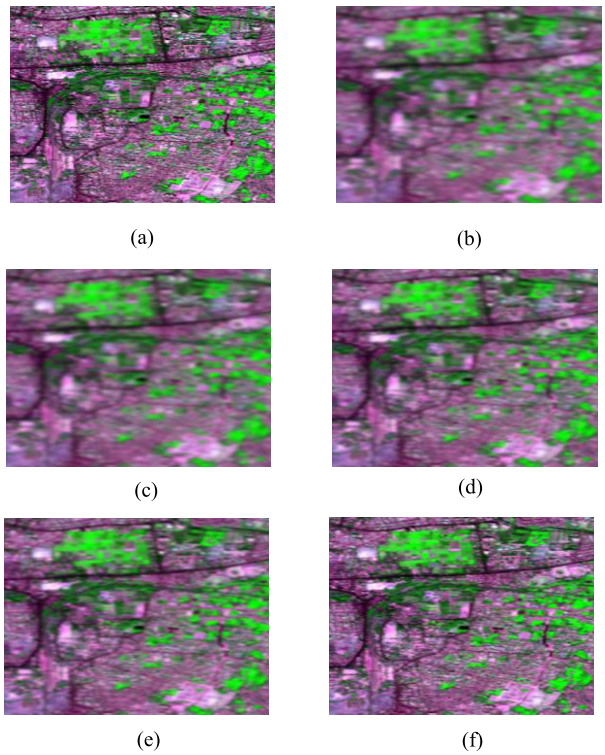
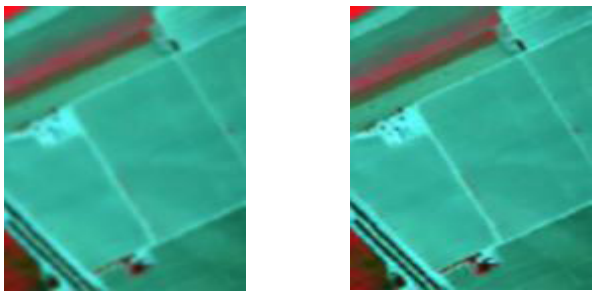


Fig. 12. Results of the proposed approach on RGB image for zoom factor = 4. (a) High-resolution image. (b) Simulated coarser version. (c) Liu *et al.* [10]. (d) Wang *et al.* [11]. (e) Mei *et al.* [18]. (f) Proposed method.

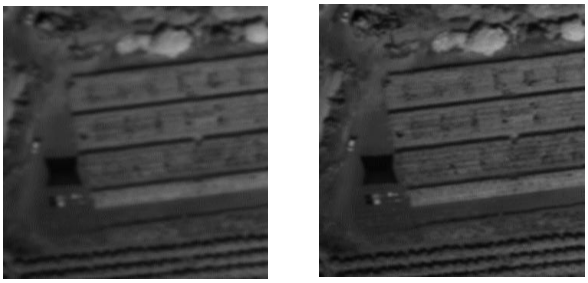
original one to assess the effectiveness of the approach. However, in order to illustrate the effectiveness of the proposed approach on real-world HSI data sets, AVIRIS-NG, UAV, and AVRIS images are employed. The visual illustrations of these experiments are summarized in Figs. 13–15. It may be noted that the enhancement is evident visually. However, due to the lack of ground truth, statistical evaluation of the

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(a) (b)

Fig. 13. Results of the proposed approach on AVIRIS data set for zoom factor = 2. (a) AVIRIS image. (b) Proposed approach.



(a) (b)

Fig. 14. Results of the proposed approach on UAV data set for zoom factor = 2. (a) UAV image. (b) Proposed approach.



(a) (b)

Fig. 15. Results of the proposed approach on AVIRIS-NG data set for zoom factor = 2. (a) AVIRIS-NG image. (b) Proposed approach.

results is not possible for the super-resolution of real-world data sets.

IV. Conclusion

Most of the prominent CNN-based SISR approaches give blurred and inaccurate results for hyperspectral data sets. Our research found that the mapping between low and high-resolution image patches, across narrow and continuous spectral bands, can be learned/generalized effectively using 3-D convolution–deconvolution layers. The initial bicubic/bilinear upscaling, adopted in earlier approaches, causes reconstruction artifacts. Hence, in our approach, before sparse coding, the coarser level hypercube is dynamically upscaled using a convolution–deconvolution network. The proposed regularized transformation, using features derived from the input hypercube, assures proper reconstruction even for inputs with a large spectral dimension. Furthermore, increase in the number of such regularizing features enhances high-frequency details in the super-resolved output, but at the cost of computational time. We also found that limiting the sparse code optimization process to a single-step convolution affects the spatial and spectral fidelity of the reconstructed images. It may be noted that the proposed implementation of ISTA as a recurrent convolutional network provides room for optimizing the sparse codes, dictionaries, and thresholds simultaneously. However, for hyperspectral super-resolution, ISTA-based strategies have limited generalization capability and also the results are found to be sensitive to the number of recurrent cycles. In order to resolve these issues, FISTA and ADMM-based approaches are proposed. Also, to overcome the bottleneck of the optimization strategies, for data with large spectral dimension, an encoder-based architecture is proposed. From the experiments over standard hyperspectral data sets, the proposed approaches are found to be better than the prominent hyperspectral SISR approaches. The proposed ADMM sub-network gives better accuracy compared to the FISTA-based strategy. Also, the former may be preferred when the number of training samples is less. However, the FITSA-based approach requires much lower number of recurrent cycles. The proposed loss functions consider the spectral characteristics of the data and are more suitable for hyperspectral data sets than their conventional counterparts. The ensemble of the proposed frameworks (where the model weights are computed based on endmember as well as regularizing feature validity) prevent the premature convergence of the models (toward local optima) in a computationally effective manner. It is found that the cascaded implementation of SISR models, for higher scale/zoom factors, is not suitable for hyperspectral images; hence the proposed approach of deconvolution should be preferred. Also, the joint unmixing-based spectral refinement of the super-resolved hypercube with respect to the coarser input, improves the generalizability of the approach. The sensitivity toward network parameter settings has been considerably reduced in our proposed approaches. Computational efficiency of the proposed models can be further improved by parallelizing the framework. The proposed architecture also seems to be promising for spectral super-resolution, which, however, needs further investigation.

Acknowledgment

Authors sincerely acknowledge Space Applications Centre (SAC), ISRO for providing the AVIRIS-NG data set.

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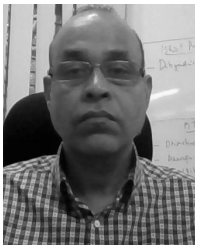


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